# Yuval Amit

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## Education

University of Maryland, College Park (UMD) ..... Aug 2023 – May 2027
B.S. in Mathematics, Minor in Computer Science (expected May 2027)

## Research Experience

SMALL REU, Williams College ..... Summer 2026
Advisor: Steven J. Miller

Undergraduate Research, UMD ..... Summer 2025
Advisors: Foivos Chnaras, Larry Washington

## Papers

*Random Fibonacci Tilings* [arXiv, 2026]
With Azmera Gebre, Christopher Housholder, Joshua Khan, Steven J. Miller, Emily Upton, Connor Yau

*Product and Quotient Set Dominance in Finite Groups* [arXiv, 2026]
With Lucas Chen, Christopher Housholder, Joshua Im, Joshua Khan, Steven J. Miller, Mateo Palomares, Devayani Pradhan

*Breaking Universality in the Lower Order Terms in the 1-level and 2-level Density of Holomorphic Cusp Newforms* [arXiv, 2026]
With Lucas Chen, Lawrence Dillon, Christopher Housholder, Xiaoyao Huang, Joshua Khan, Say-Yeon Kwon, Meiling Laurence, Steven J. Miller, Vishal Muthuvel, Devayani Pradhan, Luke Rowen, Pramana Saldin, Connor Yau, Steven Zanetti

## Reading Projects and Seminars

UMD Modular Forms and Elliptic Curves Seminar ..... Fall 2026
Text: *Introduction to Elliptic Curves and Modular Forms* (Koblitz)

*Algebraic Geometry* (Milne) ..... Fall 2026
Advisor: Harry Tamvakis

*Algebraic Number Theory and Fermat's Last Theorem* (Stewart and Tall) ..... Spring 2026
Advisor: Larry Washington

*Primes of the Form* $x^2 + ny^2$ (Cox) ..... Fall 2025
Advisor: Foivos Chnaras

*Trees* (Serre) ..... Fall 2024 – Spring 2025
Advisor: Roohollah Ebrahimian

*Topology of Numbers* (Hatcher) ..... Spring 2024
Advisor: Connor Lockhart

## Talks

UMD Math Club ..... Nov 2026
*Numbers, Groups, Graphs, Undergrad Research, and a 14-Year-Old Conjecture*

Québec-Maine Number Theory Conference ..... Oct 2026
*Breaking Universality in the Lower Order Terms in the 1-level and 2-level Density of Holomorphic Cusp Newforms*

Yale Undergraduate Conference ..... Jul 2026
*Ranks and Moments of (Hyper)elliptic Curve Families*

Combinatorial and Additive Number Theory (CANT) ..... Jul 2026
*Weighted MPTQ Sums in Finite Groups*

International Conference on Fibonacci Numbers and Their Applications ..... Jul 2026
*Random Fibonacci Tilings*

UMD Student Algebra-Number Theory ..... Nov 2025
*Genus Theory and Primes of the Form* $x^2 + ny^2$

Advanced Number Theory (Course) ..... Nov 2025
*An Introduction to the Arithmetic of Elliptic Curves*

UMD DRP Lectures ..... May 2024
*Exploring Continued Fractions with Quadratic Forms*

## Teaching and Tutoring

Co-teacher, "Proofs From the Book" student-taught course, UMD ..... Spring 2026
Tutor, Calc I & II, UMD Department of Mathematics ..... Spring 2024, Fall 2025
Tutor, High School Math, Los Altos High School AVID Program ..... Fall 2022, Spring 2023

## Awards

Dan Shanks Award, UMD Department of Mathematics ..... 2026
Aziz Mathematics Scholarship, UMD Department of Mathematics ..... 2025
President's Scholarship, University of Maryland ..... 2023
Most Promising Young Scientist, Synopsys Science Championship ..... 2023

## Service

### Conferences and Seminars

Co-organizer, UMD Lean Seminar ..... Fall 2026
Session organizer, International Conference on Fibonacci Numbers and Their Applications ..... Jul 2026

### Journals

Referee, *The Fibonacci Quarterly* ..... Summer 2026

## Skills

*Programming:* Java, C
*Math software:* Lean, SageMath
*Typesetting:* LaTeX (https://github.com/YuvalAmit2/yalatex)
*Languages:* Hebrew (conversational)